unified representations, select appropriate techniques for different scenarios, and create digital twins that understand and reason about the systems they represent.

5.1 Making sense of data

Imagine that I am not at home, but have access to the sensor data from devices that I have installed in the house. Figure 5.1 is an example of data from an air sensor stored in DynamoDB in my home digital twin. In this table, I see a collection of numbers which are raw data—the number 20.1 is an example. The headings to the table add some meaning to that raw data, so I understand that the the number 20.1 represents a temperature (but is it degrees Celsius, or Fahrenheit?).

Without any additional context around the data, I cannot make any meaningful decisions based on it. I can see that the temperature in some room of the home is around 20°, but I do not know where in my home the sensor with identifier `24e124710b423527` is currently located.

partKey *(String)*	sortKey *(Number)*	co2	humidity	pm25	presssure	temperature
24e124710b423527	1755167296031	1086	54	2	1024.1	20.1
24e124710b423527	1755167896649	988	53.5	2	1024.4	20.1
24e124710b423527	1756463299483	1430	59.5	14	1030.6	20.1
24e124710b423527	1756463899494	1525	60.5	16	1030.7	20.2
24e124710b423527	1756464499979	1222	59.5	18	1030.7	20.2

Figure 5.1 An example of raw data without context from sensors in my home digital twin. Without knowing what the data is related to, I cannot use it to make any meaningful decisions.

I could go and find all the sensors in my home and look at their identifiers until I found the sensor with the identifier of `24e124710b423527` at which point I would know that it is currently in the living room. I also happen to know that this particular sensor transmits readings in degrees Celsius, and that I had set the air conditioner in that room at 24° several hours ago, so I would expect the temperature to be higher. I also know that it has been several years since I had the unit serviced, and that the service manual recommends having the refrigerant gas replaced every two years. I can see another temperature reading in the raw data that I know relates to an outside air sensor that shows the ambient temperature to be only 5°. I'm fairly certain I remember the unit